

THE HEART



Department of anatomy

Second Part

Thorax (4)

Dr. Osman Ahmed Osman

Lecture points

➤ The Middle Mediastinum.

➤ The Pericardium.

- Fibrous pericardium.
- Serous Pericardium.
- Pericardial Cavity.
- Pericardial Sinuses.
- Nerve supply.
- Blood supply.
- Pericardiocentesis.

➤ Location of the heart .

➤ Cardiac Orientation.

➤ Base of the heart.

➤ Anterior surface of the heart.

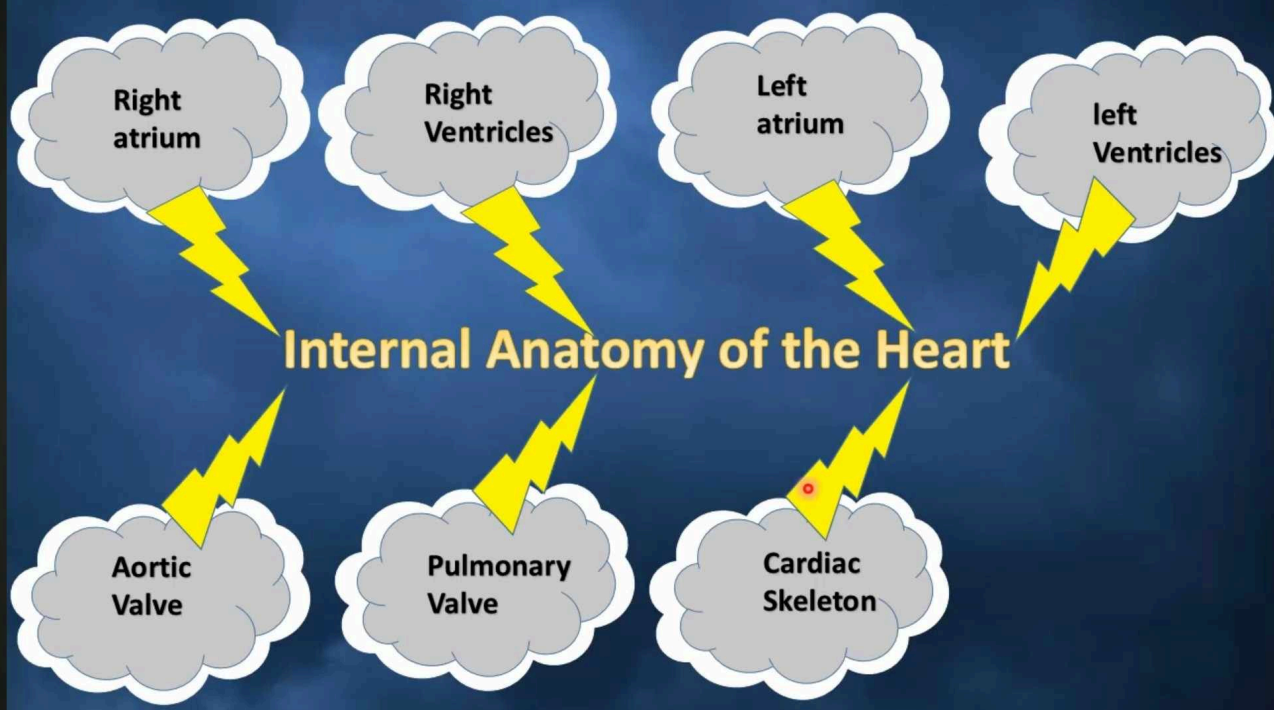
➤ Diaphragmatic surface.

➤ The Border and margin of the heart.

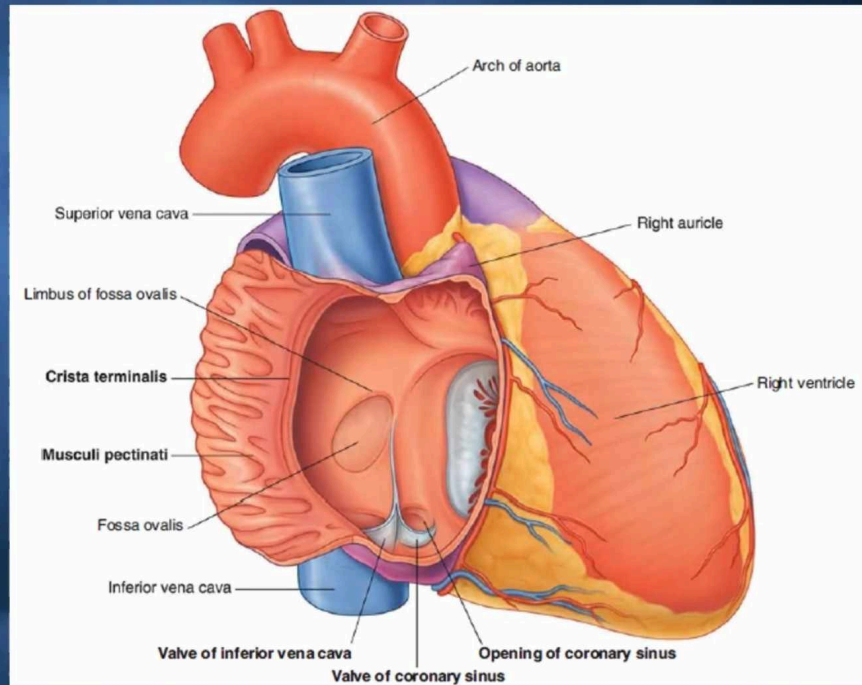
➤ External Sulci (groove) .

➤ Internal Anatomy of the Heart

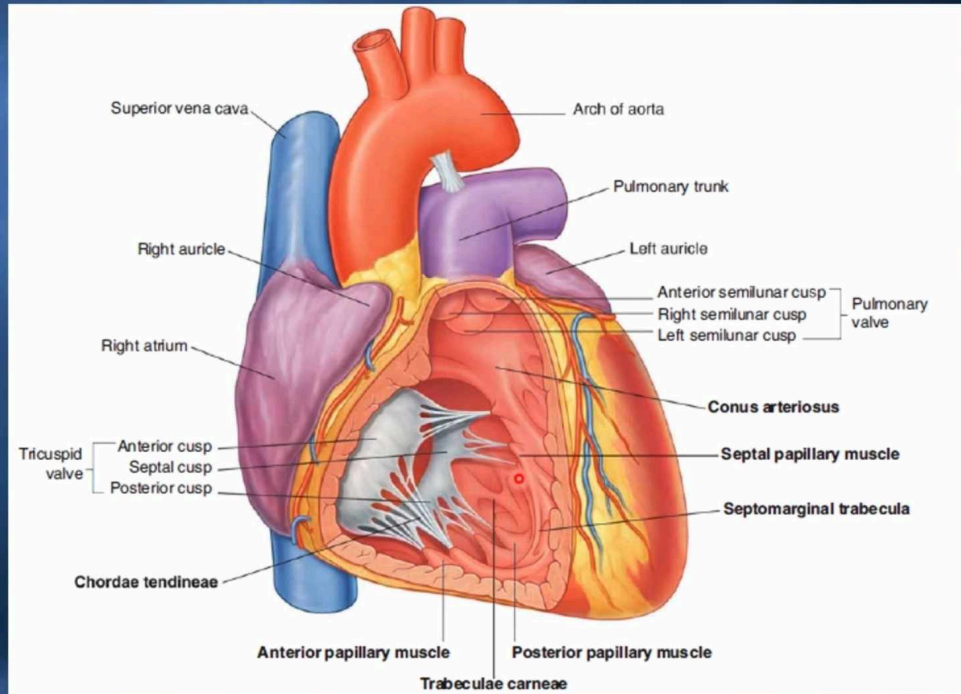
- Right atrium.
- Right Ventricles.
- Left atrium.
- left Ventricles.
- Aortic Valve .
- Pulmonary Valve .
- Cardiac Skeleton.
- Coronary Arteries.
 - Right Coronary Arteries.
 - Left Coronary Arteries.
 - Coronary Arteries Variation.
- Coronary Sinus.
- Conductive system of the heart.
- Cardiac Plexus
- Passage of Blood Through the Heart.
- location of the heart valves, and auscultation points.



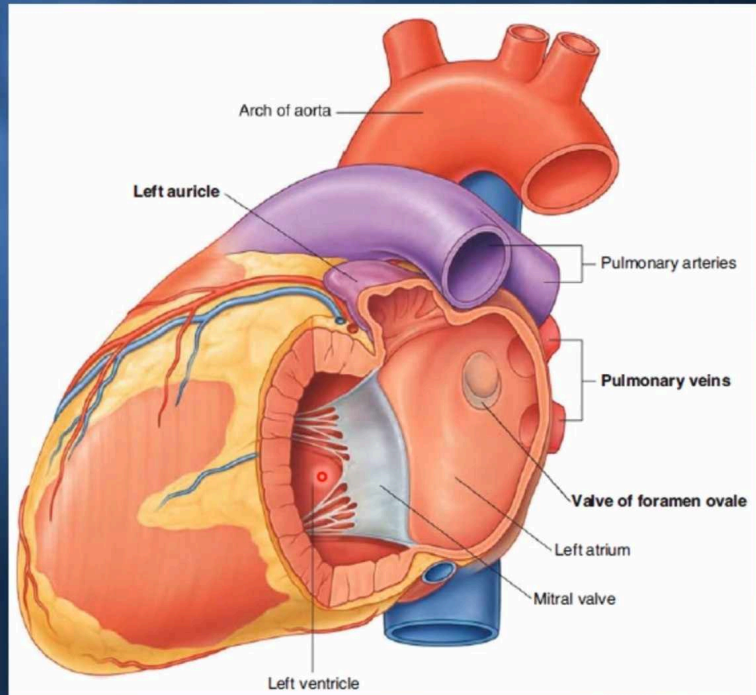
Right atrium



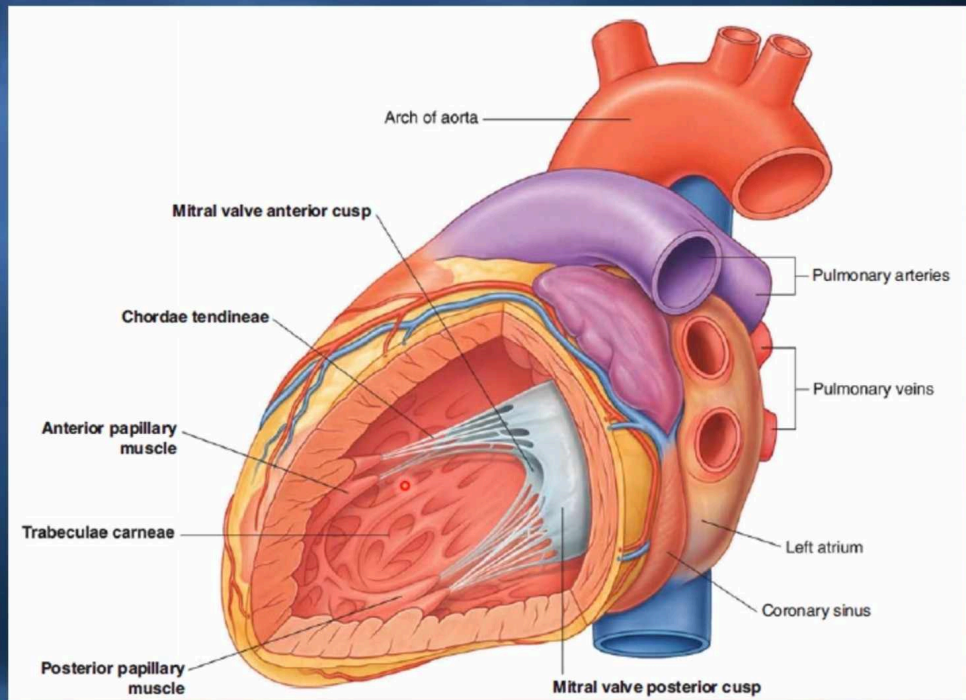
Right Ventricles



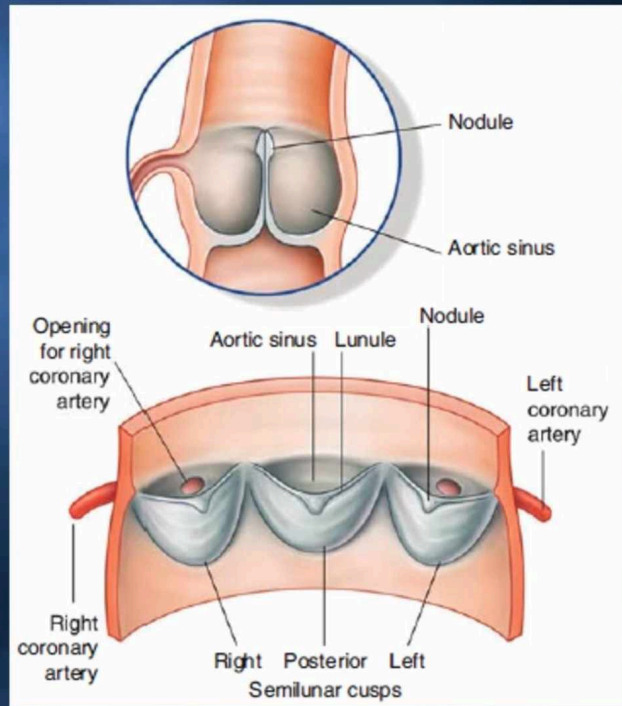
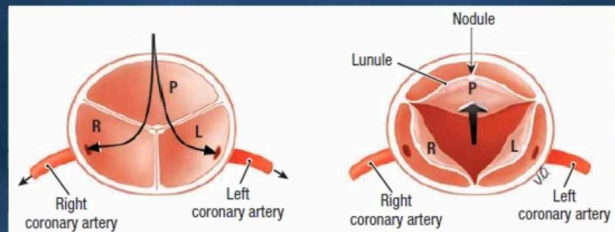
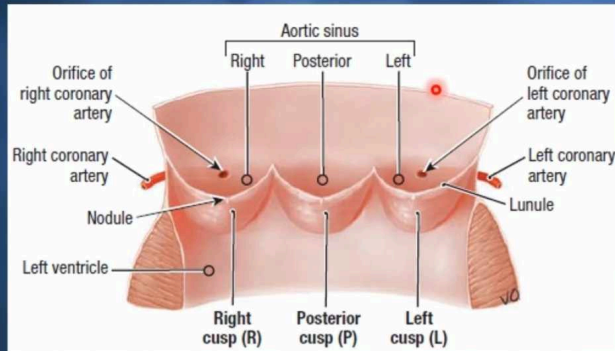
Left atrium



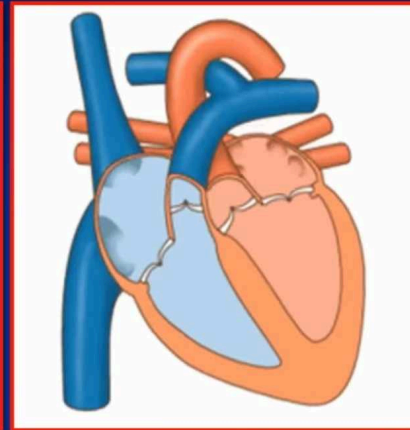
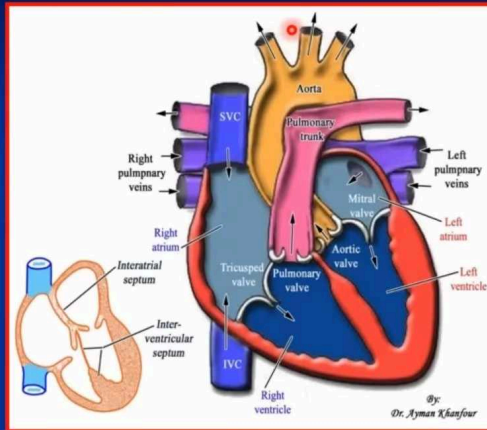
Left Ventricles



Aortic Valve

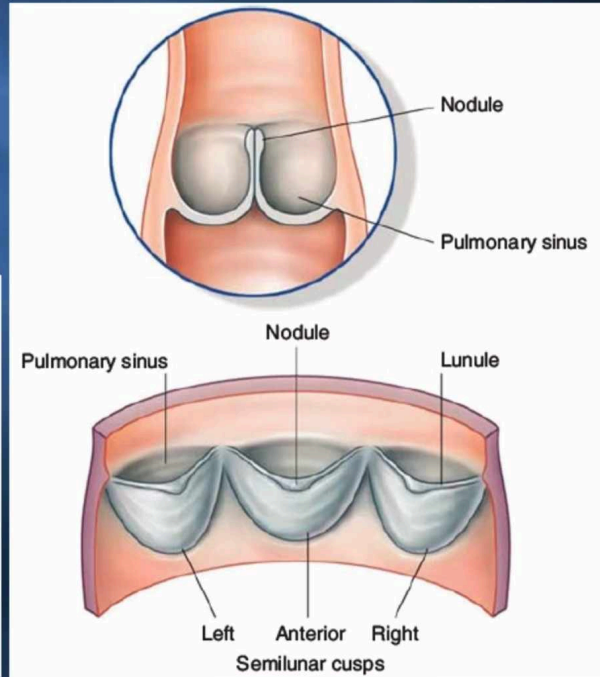
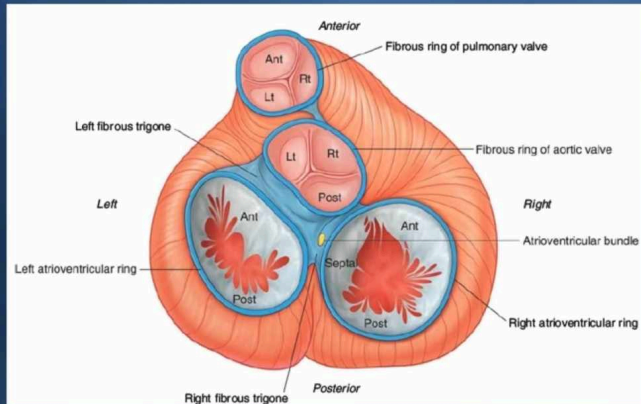


❖ How blood circulates inside the heart?



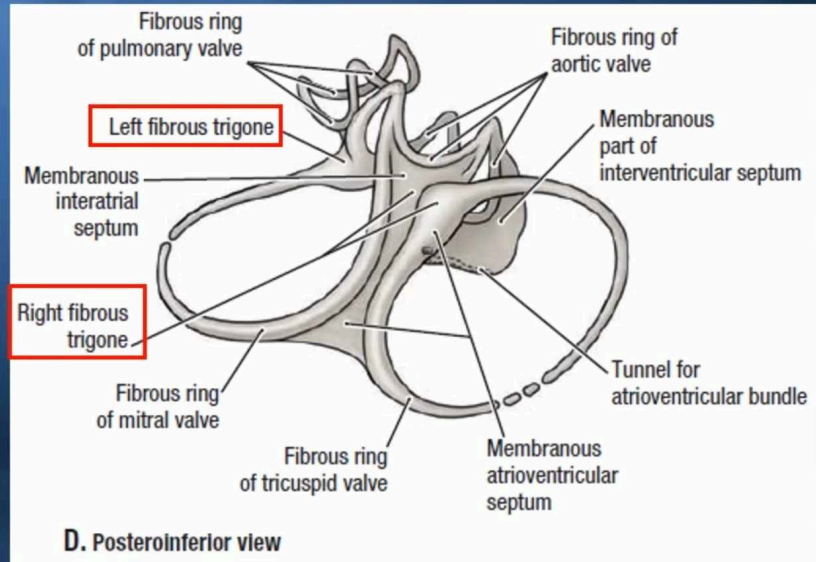
(Chambers, valves and great vessels of the heart)

Pulmonary Valve

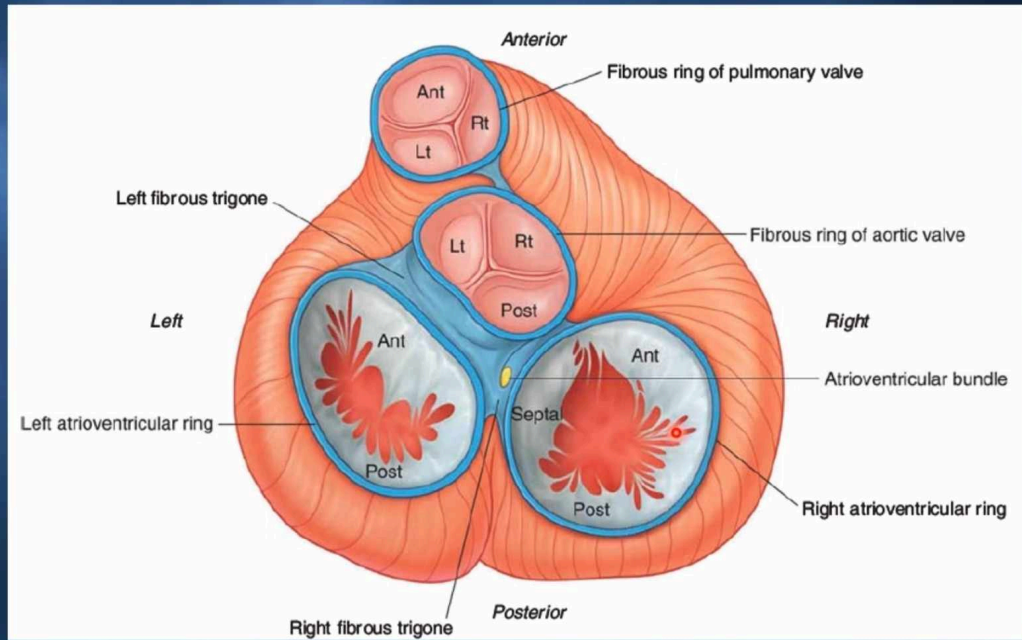


Cardiac Skeleton

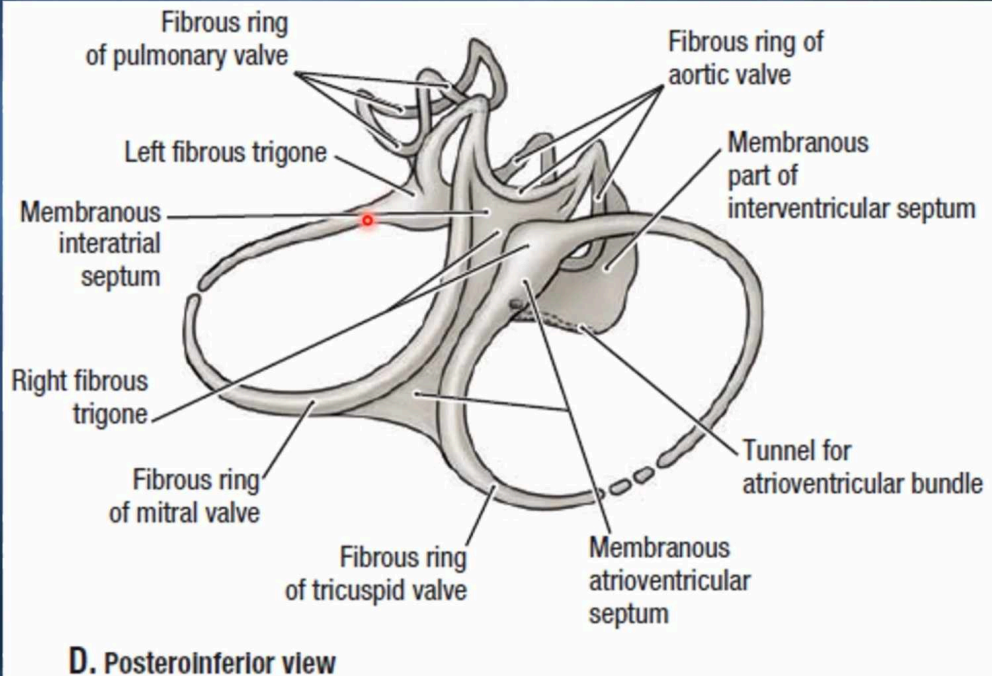
- It is collection of dense, **fibrous connective tissue** in the form of **four rings** with **interconnecting areas** in plane between the atria and the ventricles.
- The four rings of the cardiac skeleton surround **the two atrioventricular orifices, the aortic orifice and opening of the pulmonary artery**.
- They are **the anulus fibrosus** as the interconnecting areas include: (**Right fibrous trigone & Left fibrous trigone**).



Cardiac Skeleton

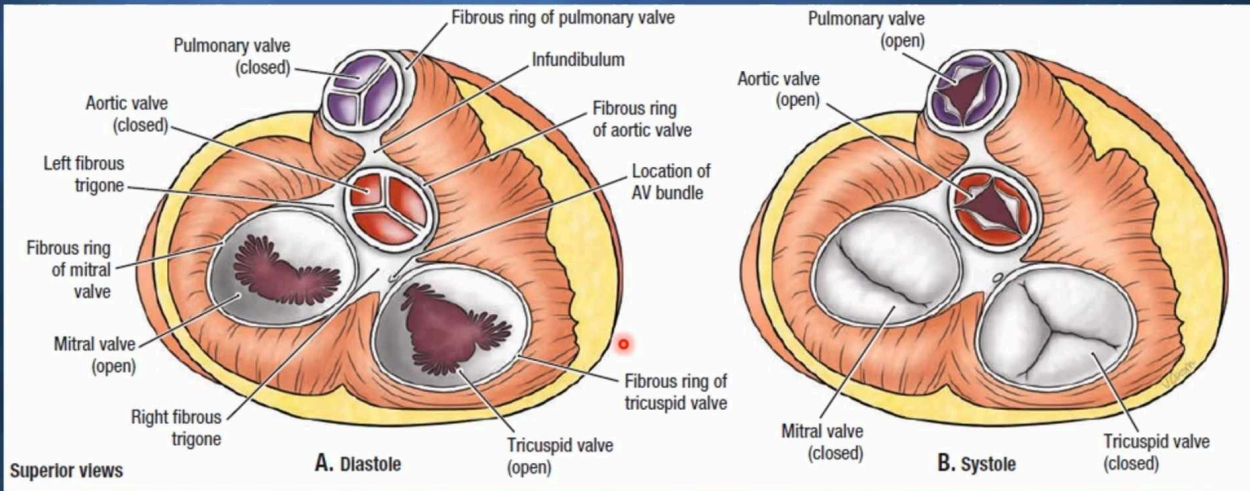


Cardiac Skeleton



Cardiac Skeleton

- It keep the orifice of the valve patent
- It separate myoentric impulses of the atrium
- It provide attachment of the leaflete and cusp of the valve.

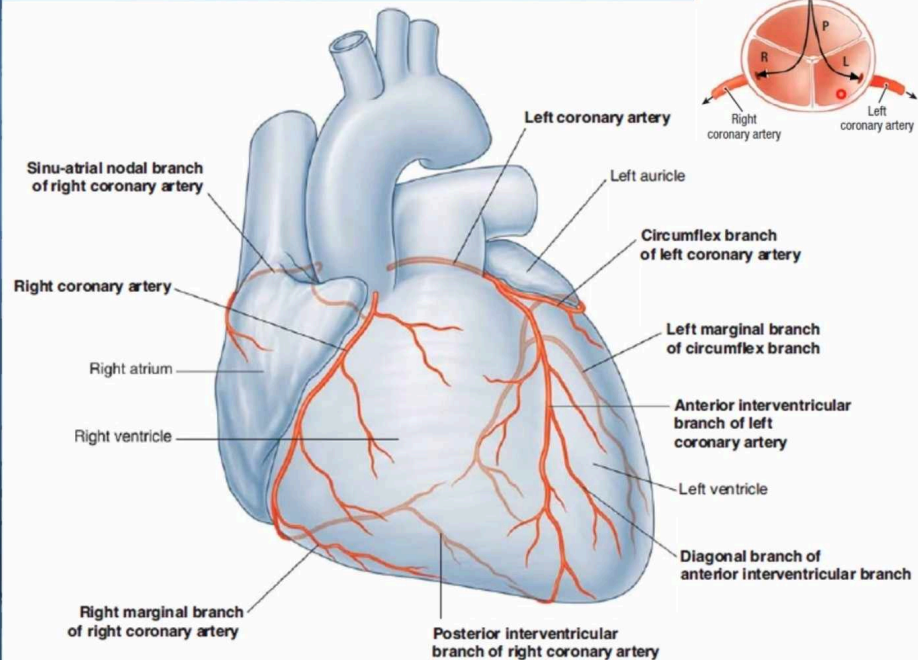


Coronary Arteries

- They arise from the ascending aorta.
- They are filled with blood during the ventricular diastole.

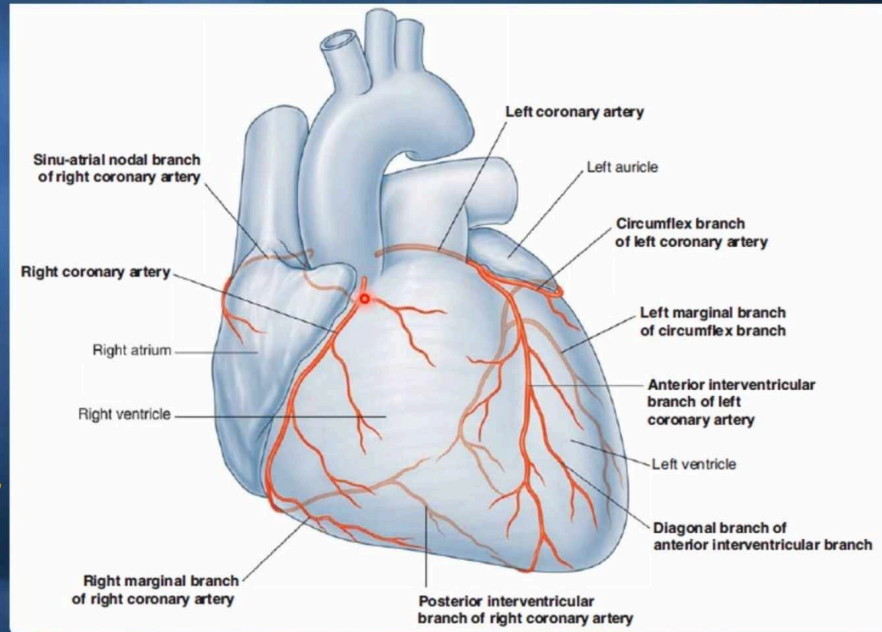
1. Right Coronary Artery.

2. Left Coronary Artery.



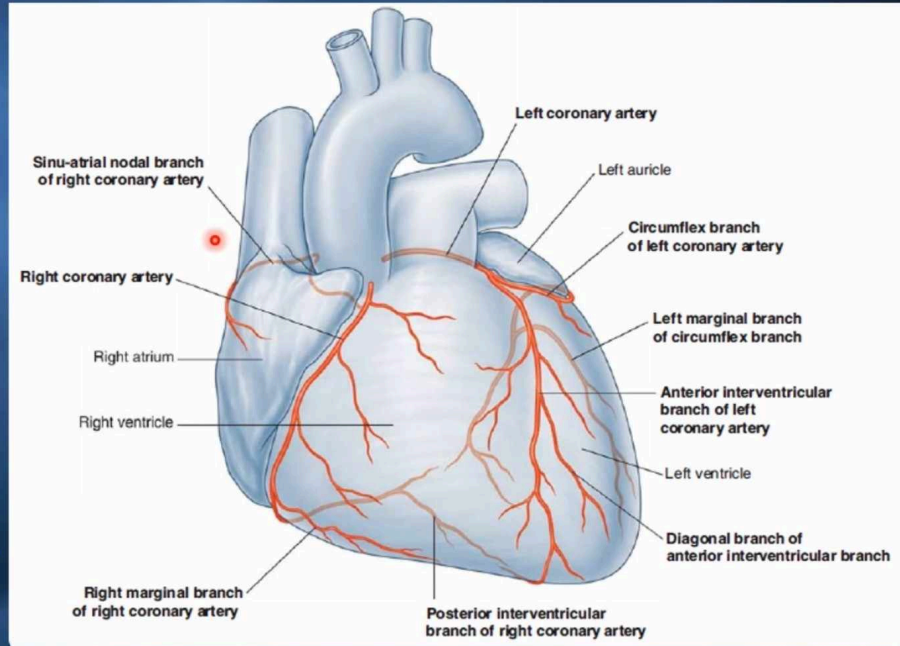
Right Coronary artery

- It arises from the anterior (right) aortic sinus of the ascending aorta.
- It gives rise to the following:
 - **SA Nodal Artery.**
 - **Marginal Artery.**
 - **Posterior IV (Posterior Descending) Artery.**
 - **AV Nodal Artery.**

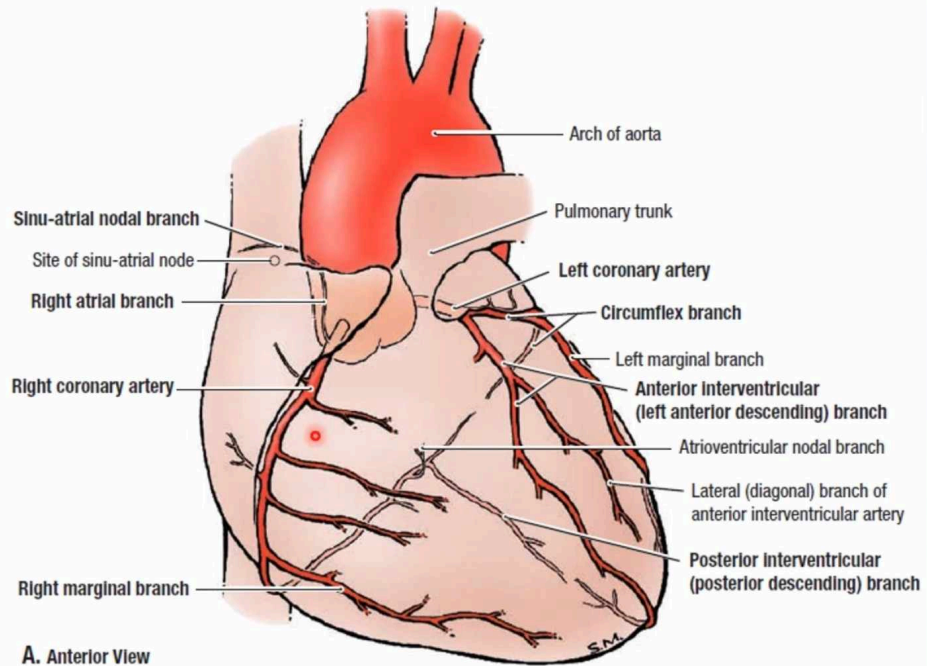


Left coronary artery

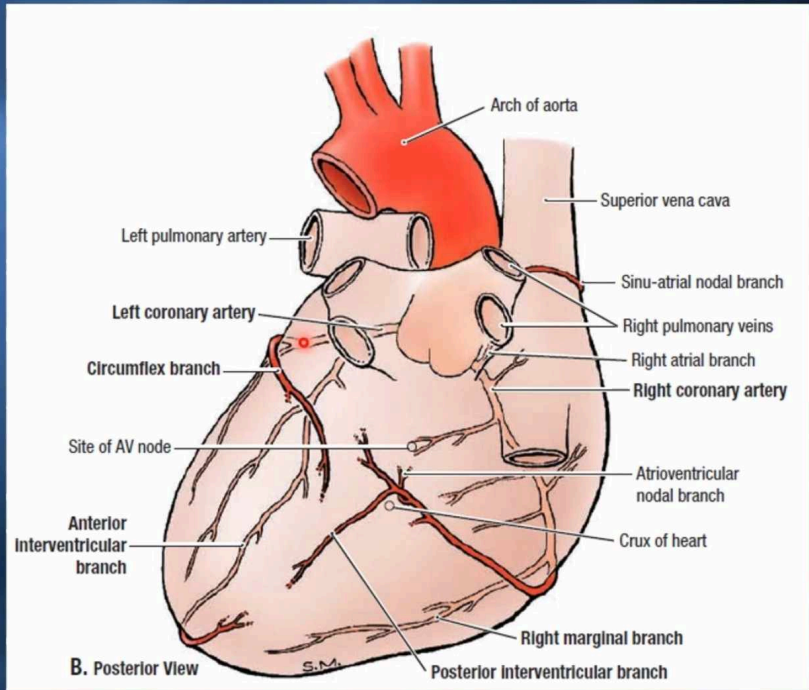
- It arises from the left aortic sinus of the ascending aorta, just above the aortic semilunar valve.
- It is shorter than the right coronary artery.
- It gives rise to the following:
 - **The anterior IV (Left Anterior Descending) artery.**
 - **The circumflex artery.**



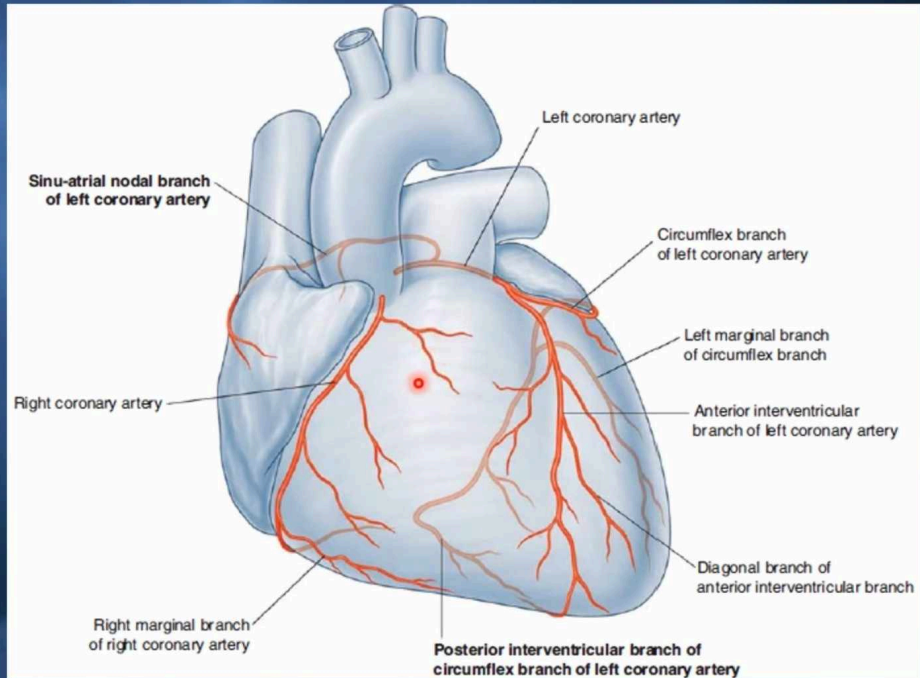
Coronary Arteries



Coronary Arteries

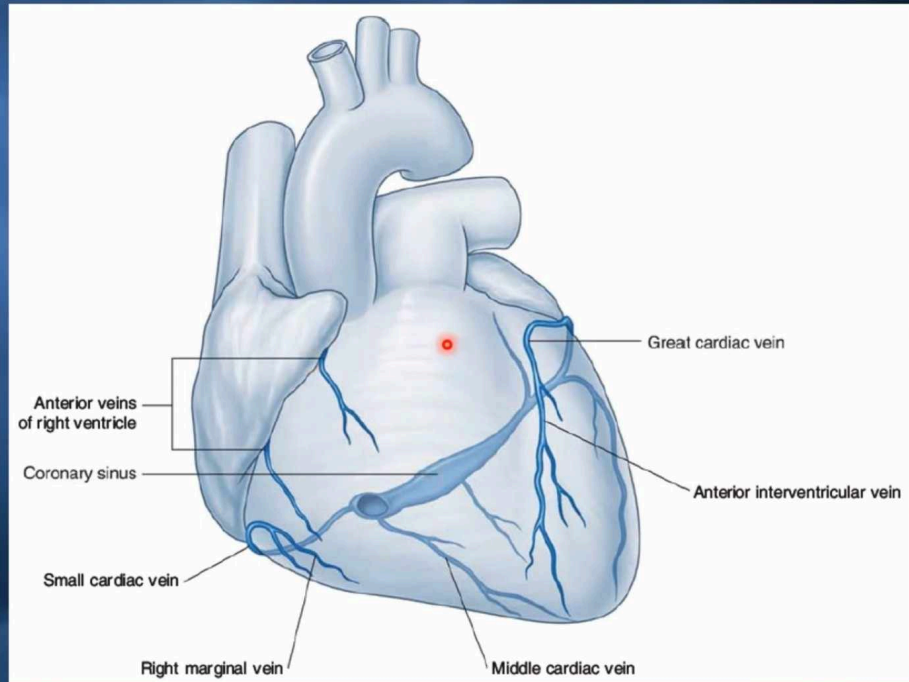


Coronary Arteries Variation



Coronary Sinus

- It is the **largest vein draining the heart.**
- It lies in the **coronary sulcus**, which separates the atria from the ventricles.
- It **opens into the right atrium.**

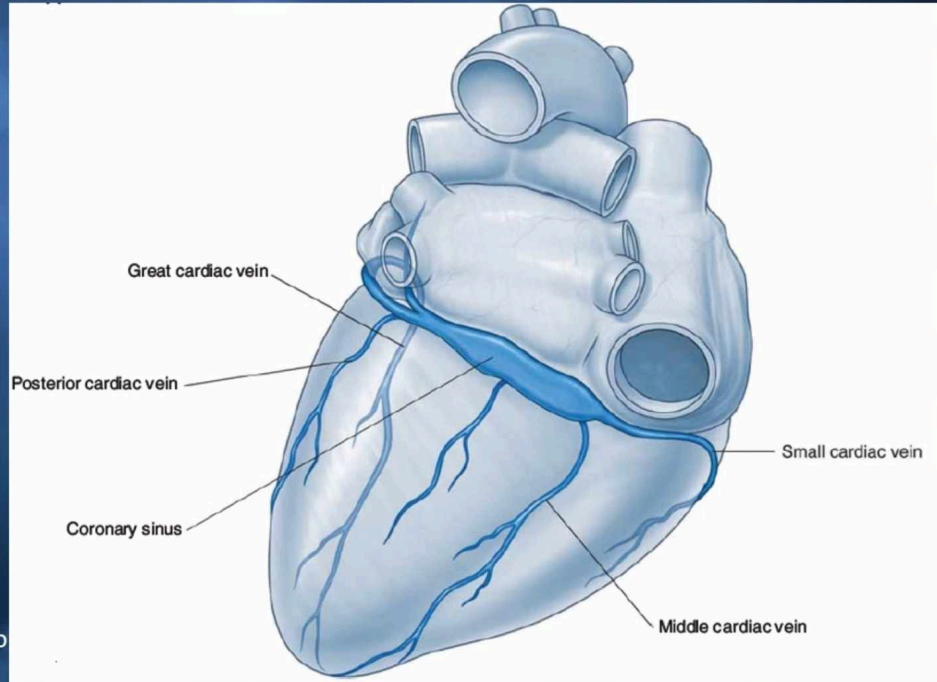


Coronary sinus

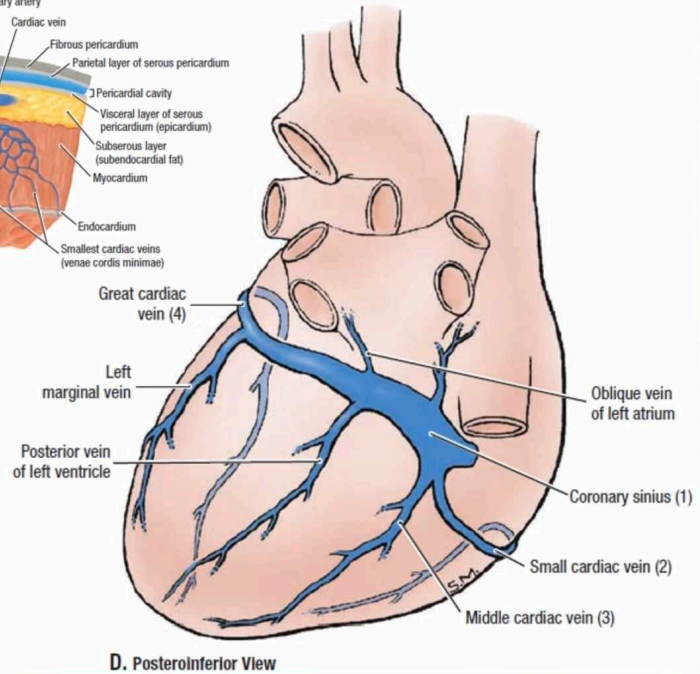
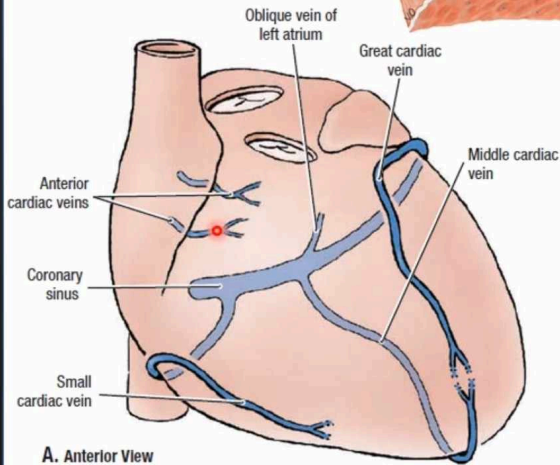
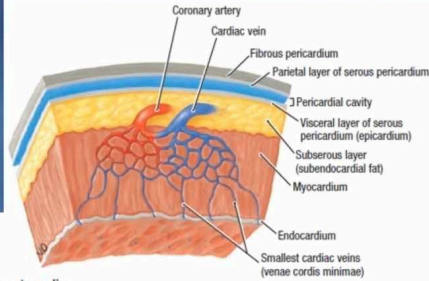
It receives:

- 1) Anterior interventricular vein (drain in great Cardiac vein).
- 2) Great Cardiac Vein.
- 3) Posterior cardiac vein.
- 4) Middle Cardiac Vein.
- 5) Small Cardiac Vein.
- 6) Oblique Vein of the Left Atrium.
- 7) **Smallest Cardiac Veins** (Venae Cordis Minimae).

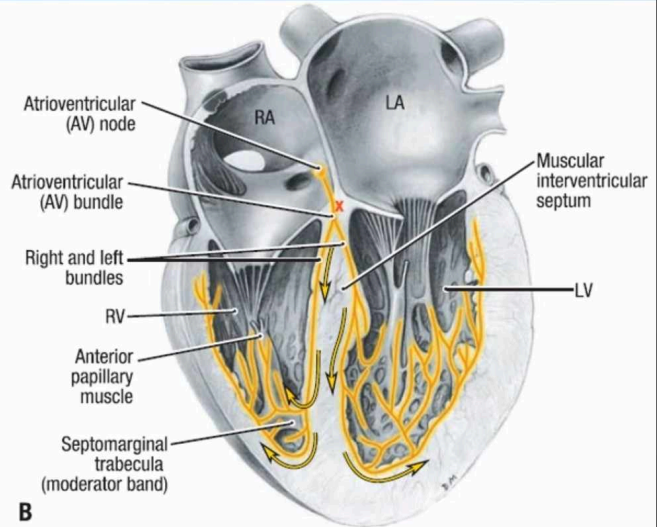
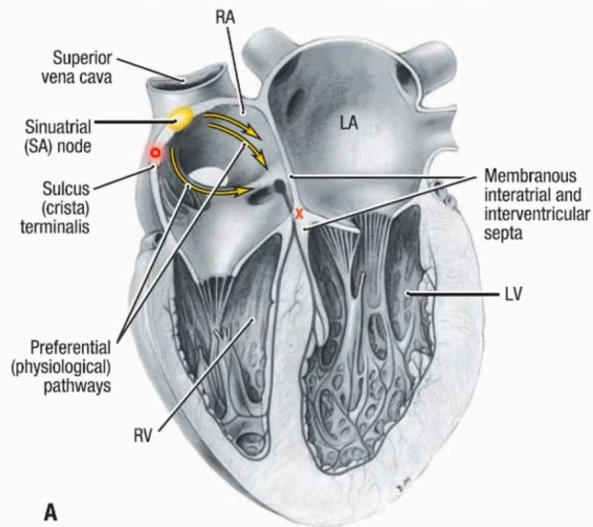
(Begin in the wall of the heart and empty directly into its chambers).



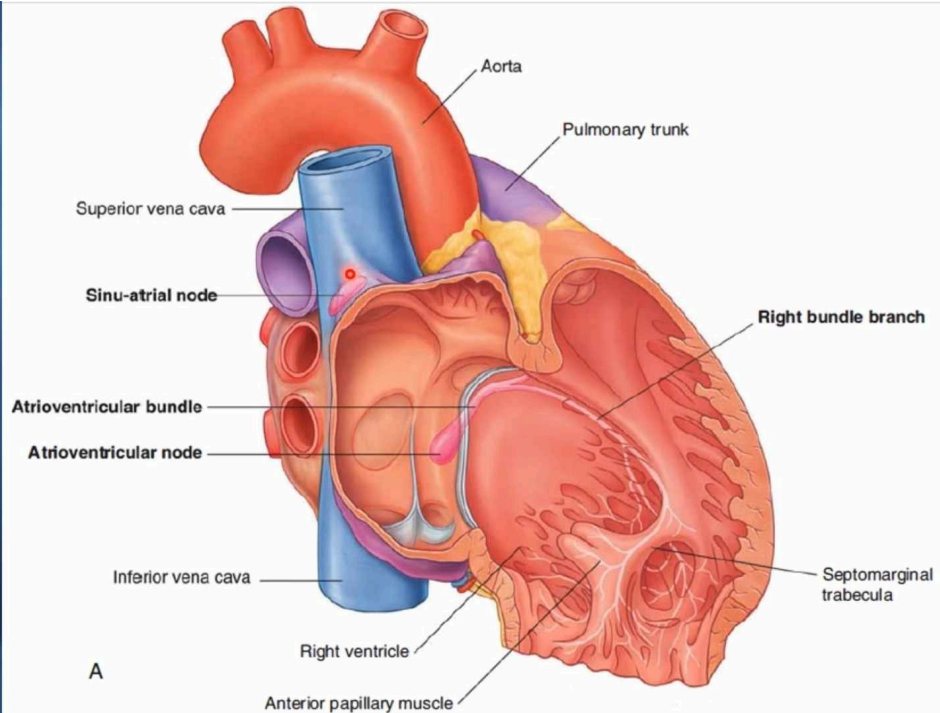
Coronary sinus



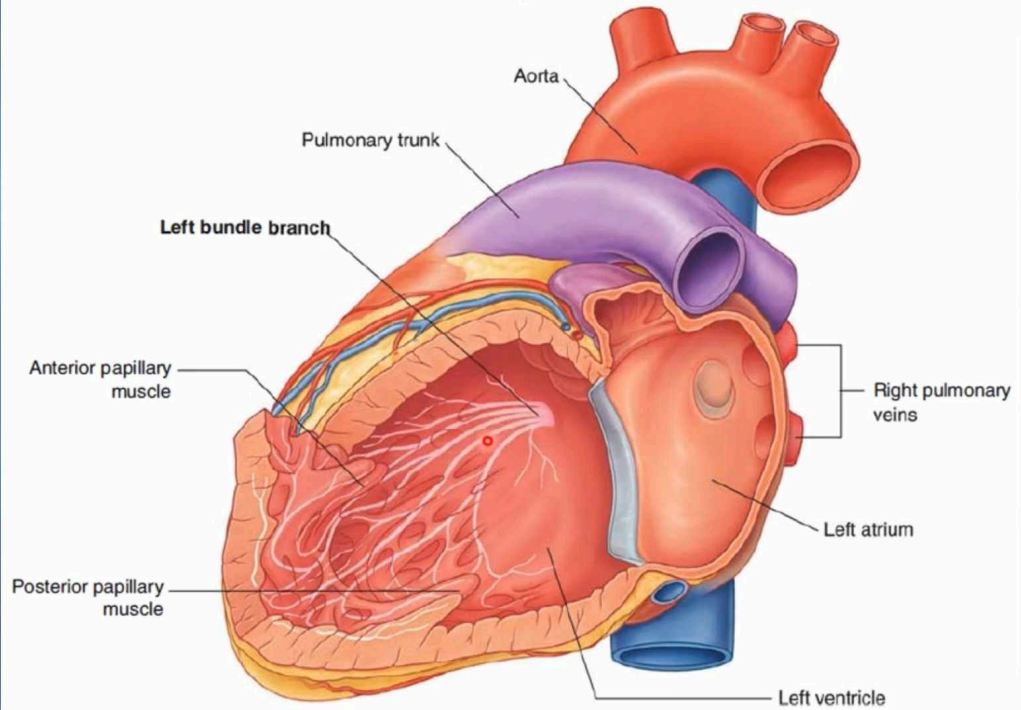
Conductive system of the heart



Conductive system of the heart



Conductive system of the heart



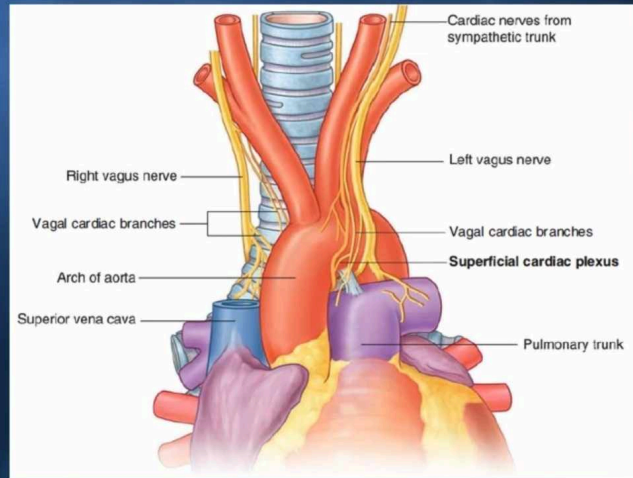
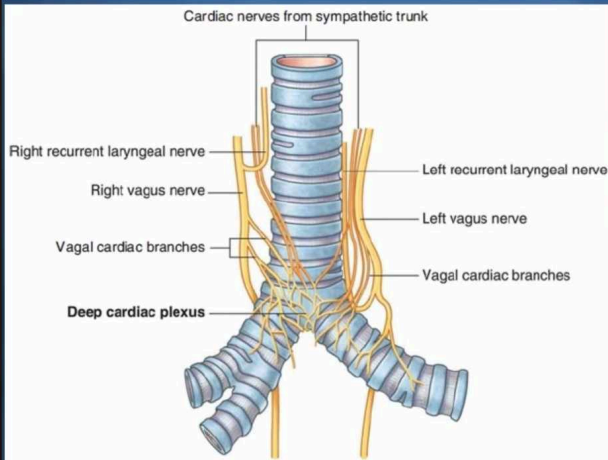
Cardiac Plexus

It receives the superior, middle, and inferior cervical and thoracic cardiac nerves from the sympathetic trunks and vagus nerves.

It is divisible into:

- The superficial cardiac plexus.
- The deep cardiac plexus.

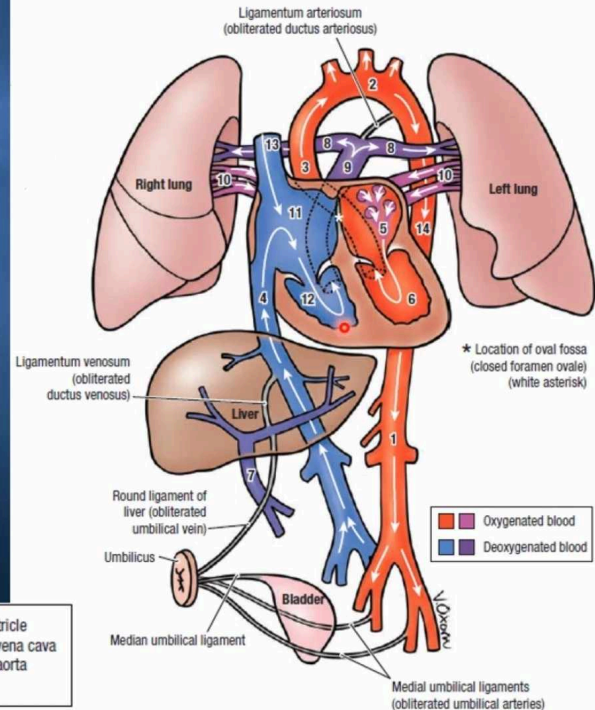
It innervates the conducting system of the heart: **sympathetic** and **parasympathetic** branches.

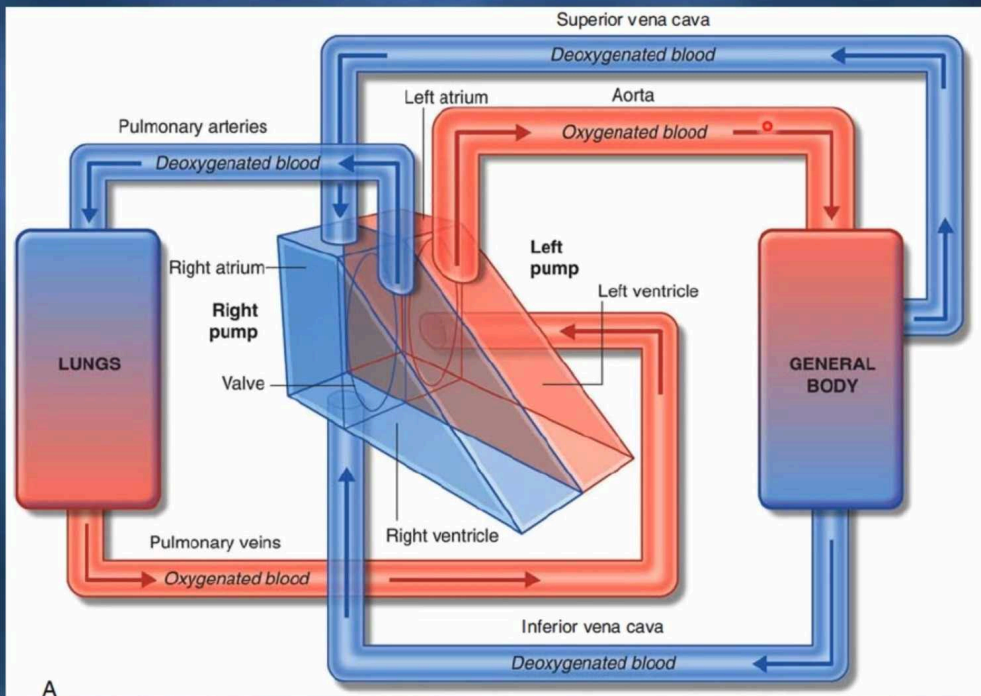


Passage of Blood Through the Heart

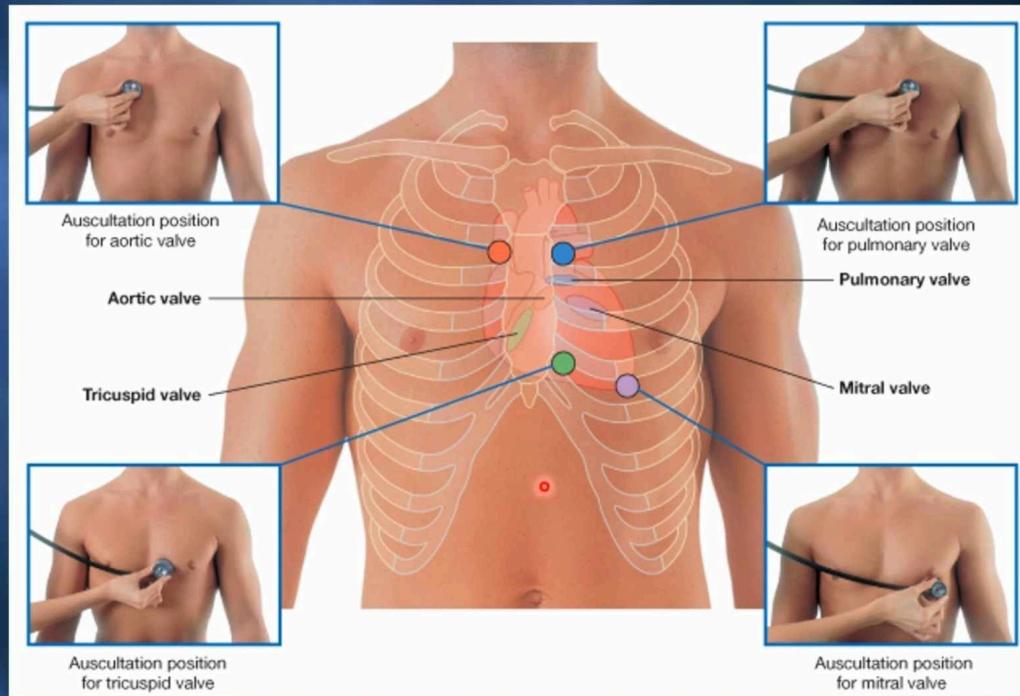
- Blood follows this sequence through the heart:
- superior and inferior vena cava → right atrium → tricuspid valve → right ventricle → pulmonary semilunar valve → pulmonary trunk and arteries to the lungs → pulmonary veins leaving the lungs → left atrium → bicuspid valve → left ventricle → aortic semilunar valve → aorta → to the body.

Heart and blood vessels:	4 Inferior vena cava	8 Pulmonary arteries	12 Right ventricle
1 Abdominal aorta	5 Left atrium	9 Pulmonary trunk	13 Superior vena cava
2 Arch of aorta	6 Left ventricle	10 Pulmonary veins	14 Thoracic aorta
3 Ascending aorta	7 Portal vein	11 Right atrium	





**Anterior view
of the chest wall of a man
showing
location of
the heart
valves, and
auscultation
points**



Thanks